Chinese Character Style Transfer with Conditional GAN

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Abstract

The purpose of this paper is to transfer the style of Chinese characters from standard font stylized words to multiple styles by using Conditional generative adversarial network. This research is expected to open the door to enrich the stylization of pictographs, not only for Chinese characters, but also for other pictographs in the future, such as Japanese and Korean, other than alphabet stylization, to fill the gap in this field. To achieve this, the research used CGAN, with methods such as U-Net to construct the basic model framework. In the research, some new ideas were proposed based on pix2pix, adding category loss and constant loss to the network structure, making the output -results more idealized. And the final work achieved the expected results, outputting Chinese characters in different styles and inferring the same shapes as the basic facts, successfully handling more stylized and complex fonts. This means that the Conditional generative adversarial network can handle the Chinese character style problem very well, and more types of research are expected to be solved by this standard in the future.

1 Introduction

This paper aims to show how to transfer Chinese characters into different styles by using Conditional generative adversarial network to solve the current situation of single, lacking, and illegible Chinese character stylization, and to enrich the diversity of Chinese character styles to suit various needs. There are many similar studies in the field of Chinese character stylization, but the results are unsatisfactory, as the generated text images are usually blurred and only simple stylization can be performed. In this research, GAN is applied to font learning based on the pix2pix network structure, and the training results are optimized by adding two kinds of losses, category loss and constant loss, to obtain better training results. The experimental results are based on two methodologies, CGAN and U-Net.

This paper contains five main sections, introduction, methodology, results and conclusion, with references at the end of the paper. In the methodology section, the two methodologies, CGAN and U-Net, are introduced. The model of the project, category Embedding, and some special losses are introduced and visualized in the form of images. In the results section, the results of the study are presented in the form of images. In the conclusion section, the results of the study are mentioned, the flaws of the research are analyzed and its implications for future research.

2 Methods

2.1 CGAN

GAN was used to learn a loss that tries to classify if the output image is real or fake, while simultaneously training a generative model to minimize this loss. The generator G is trained to produce outputs that cannot be distinguished from "real" images by an adversarially trained discriminator, D, which is trained to do as well as possible at detecting the generator's "fakes".

When it comes to CGAN, it was explored in the conditional setting. Just as GANs learn a generative model of data, conditional GANs (cGANs) learn a conditional generative model. We condition on an input image and generate a corresponding output image.

In other words, GANs are generative models that learn a mapping from random noise vector z to output image y, $G : z \rightarrow y$. In contrast, conditional GANs learn a mapping from observed image x and random noise vector z, to y, $G : x, z \rightarrow y$.

2.2 U-Net

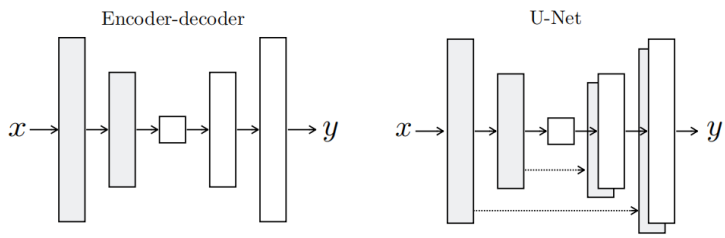

Figure 1: U-NET

In the traditional encoder-decoder network, the input is passed through a series of layers that progressively downsample, until a bottleneck layer, at which point the process is reversed. Such a network requires that all information flow pass through all the layers, including the bottleneck.

U-net is added the skip connection, as there is a great deal of low-level information shared between the input and output, and it would be desirable to shuttle this information directly across the net.

Specifically, we add skip connections between each layer i and layer n i, where n is the total number of layers. Each skip connection simply concatenates all channels at layer i with those at layer n i.

2.3 Category Embedding

The introduction of category embedding is concatenating a non-trainable Gaussian noise as style embedding to the character embedding, right before it goes through decoder.

By having this, the encoder still maps same character into the same vector. But the decoder will take both the character and the category embedding to generate the target character.

2.4 Overall model

Both generator and discriminator use modules of the form convolution-BatchNorm-ReLu.

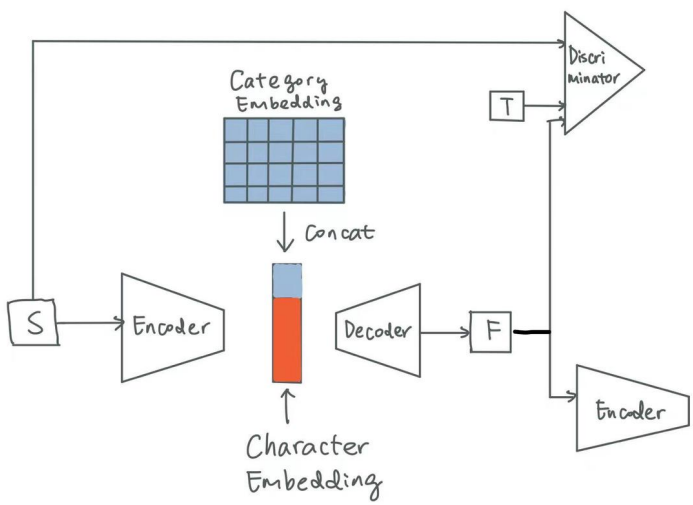

Figure 2: Model Structure

2.5 Loss funtion

Overall, the loss function contains the generator loss function and discriminator loss function.

$$loss_g = cheatLoss + L1Loss + fakeCategoryLoss + constantLoss + tvLoss$$

$$loss_d = drealLss + dfakeLoss + 0.5 * fakeCategoryLoss$$

To be more detailed, the category loss makes the discriminator to penalize when the font is not the willing target fonts. Constant loss lets the source and generated character resemble the same character information, having the same language meaning.

3 Results

Figure 3: reference results

The results given in the Figure3 above is satisfactory as a whole.For majority of the fonts, the network succeeds to make reasonable inferences. Some of them are actually very close to the ground truth. Also worth noting, the tiny but distinguishable details are preserved by the network, like the curly end of radicals.

So the outcomes are quite exciting because we are able to enrich the diversity of Chinese character style to suit various needs in designing the fonts from these single, original and lacking-aesthetic feelings Chinese to the distinct beautiful fonts.

But just like many other neural network powered applications, when the network fails, it fails spectacularly. For some fonts, especially the ones that are light in weight, it merely comes up some blurry inky blob. On the other hand, for those heavy fonts, it loses critical details of blanks to make the character distinguishable, only capturing the overall outline. While for some characters, there will be some ambiguity and distortion. Through carefully checking, it is found that the unsatisfying results happened when the target font and source font have a big difference. This will also be the future research direction for us.

4 Conclusion

This goal of the paper is to generate different Chinese fonts by transferring the original style of the font to the other characters using the conditional generated adversarial network.

The comparison with ground truth is shown above(Figure3) , with left being the original character and right as the inferred characters. As we can compare the generated characters with the ground truth, for most characters, the inferred shape is almost identical to its ground truth.But still there are some words which not perform very well because they are cursive or chubby in some certain ways.

In the future, we would love to see that we train a model to learn the typefaces of all the languages in the world, that later on, people just need to design a single font, then automatically get the corresponding fonts in all the other languages.While to be more concrete, we would like to explore creating different fonts with mixed styles.And we want to also try different GANs like StackGAN, better GAN models like WGAN and LSGAN(Loss Sensitive GAN), and other domain transfer network like DiscoGAN with it .We think that they may produce more accurate outcomes.

Last but not least, this project demonstrates a possibility of applying deep learning, more specifically, GAN in helping accelerating design process for Chinese fonts. The results are promising and encouraging and maybe in the future it will magically create new font out of box.

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